

Answer ALL questions. Write your answers in the spaces provided.

1. A chocolate manufacturer places special tokens in 2% of the bars it produces so that each bar contains at most one token. Anyone who collects 3 of these tokens can claim a prize.

Andreia buys a box of 40 bars of the chocolate.

- (a) Find the probability that Andreia can claim a prize.

(2)

Barney intends to buy bars of the chocolate, one at a time, until he can claim a prize.

- (b) Find the probability that Barney can claim a prize when he buys his 40th bar of chocolate.

(3)

- (c) Find the expected number of bars that Barney must buy to claim a prize.

(1)

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7. A spinner can land on red or blue. When the spinner is spun, there is a probability of $\frac{1}{3}$ that it lands on blue. The spinner is spun repeatedly.

The random variable B represents the number of the spin when the spinner first lands on blue.

(a) Find (i) $P(B = 4)$

(ii) $P(B \leq 5)$

(4)

(b) Find $E(B^2)$

(3)

Steve invites Tamara to play a game with this spinner.

Tamara must choose a colour, either red or blue.

Steve will spin the spinner repeatedly until the spinner first lands on the colour Tamara has chosen. The random variable X represents the number of the spin when this occurs.

If Tamara chooses red, her score is e^X

If Tamara chooses blue, her score is X^2

- (c) State, giving your reasons and showing any calculations you have made, which colour you would recommend that Tamara chooses.

(5)

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3. Suzanne and Jon are playing a game.

They put 4 red counters and 1 blue counter in a bag.

Suzanne reaches into the bag and selects one of the counters at random. If the counter she selects is blue, she wins the game. Otherwise she puts it back in the bag and Jon selects one at random. If the counter he selects is blue, he wins the game. Otherwise he puts it back in the bag and they repeat this process until one of them selects the blue counter.

- (a) Find the probability that Suzanne selects the blue counter on her 4th selection. (2)
- (b) Find the probability that the blue counter is first selected on or after Jon's third selection. (2)
- (c) Find the mean and standard deviation of the number of selections made until the blue counter is selected. (2)
- (d) Find the probability that Suzanne wins the game. (3)

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5. Asha, Davinda and Jerry each have a bag containing a large number of counters, some of which are white and the rest are red.
Each person draws counters from their bag one at a time, notes the colour of the counter and returns it to their bag.

The probability of Asha getting a red counter on any one draw is 0.07

- (a) Find the probability that Asha will draw at least 3 white counters before a red counter is drawn. (2)
- (b) Find the probability that Asha gets a red counter for the second time on her 9th draw. (2)

The probability of Davinda getting a red counter on any one draw is p .
Davinda draws counters until she gets n red counters. The random variable D is the number of counters Davinda draws.

Given that the mean and the standard deviation of D are 4400 and 660 respectively,

- (c) find the value of p . (4)

Jerry believes that his bag contains a smaller proportion of red counters than Asha's bag. To test his belief, Jerry draws counters from his bag until he gets a red counter. Jerry defines the random variable J to be the number of counters drawn up to and including the first red counter.

- (d) Stating your hypotheses clearly and using a 10% level of significance, find the critical region for this test. (5)

Jerry gets a red counter for the first time on his 34th draw.

- (e) Giving a reason for your answer, state whether or not there is evidence that Jerry's bag contains a smaller proportion of red counters than Asha's bag. (2)

Given that the probability of Jerry getting a red counter on any one draw is 0.011

- (f) show that the power of the test is 0.702 to 3 significant figures. (3)



- (4)

(3)

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5. A random sample of 150 observations is taken from a geometric distribution with parameter 0.3

Estimate the probability that the mean of the sample is less than 3.45

(5)

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4. There are 32 students in a class.

Each student rolls a fair die repeatedly, stopping when their total number of sixes is 4
Each student records the total number of times they rolled the die.

Estimate the probability that the mean number of rolls for the class is less than 27.2

(6)

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7. Each time a spinner is spun, the probability that it lands on red is 0.2

(a) Find the probability that the spinner lands on red

(i) for the 1st time on the 4th spin

(2)

(ii) for the 3rd time on the 8th spin

(2)

(iii) exactly 4 times during 10 spins

(2)

Each time the spinner is spun, the probability that it lands on yellow is 0.4

In a game with this spinner, a player must choose one of two events

R is the event that the spinner lands on **red** for the **1st** time in at most 4 spins

Y is the event that the spinner lands on **yellow** for the **3rd** time in at most 7 spins

(b) Showing your calculations clearly, determine which of these events has the greater probability.

(7)

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7. The probability of winning a prize when playing a single game of *Pento* is $\frac{1}{5}$

When more than one game is played the games are independent.
Sam plays 20 games.

- (a) Find the probability that Sam wins 4 or more prizes. (2)

Tessa plays a series of games.

- (b) Find the probability that Tessa wins her 4th prize on her 20th game. (2)

Rama invites Sam and Tessa to play some new games of *Pento*.

They must pay Rama £1 for each game they play but Rama will pay them £2 for the first time they win a prize, £4 for the second time and £(2w) when they win their wth prize ($w > 2$)

Sam decides to play n games of *Pento* with Rama.

- (c) Show that Sam's expected profit is $\pounds \frac{1}{25}(n^2 - 16n)$ (6)

Given that Sam chose $n = 15$

- (d) find the probability that Sam does not make a loss. (4)

Tessa agrees to play *Pento* with Rama. She will play games until she wins r prizes and then she will stop.

- (e) Find, in terms of r , Tessa's expected profit. (4)

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5. The probability of Richard winning a prize in a game at the fair is 0.15

Richard plays a number of games.

- (a) Find the probability of Richard winning his second prize on his 8th game, (2)

- (b) State two assumptions that have to be made, for the model used in part (a) to be valid. (2)

Mary plays the same game, but has a different probability of winning a prize. She plays until she has won r prizes. The random variable G represents the total number of games Mary plays.

- (c) Given that the mean and standard deviation of G are 18 and 6 respectively, determine whether Richard or Mary has the greater probability of winning a prize in a game. (4)

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